MTE 203 – Advanced Calculus
Week 5 Homework

Higher Order Partial Derivatives and Applications (S.12.5)

Problem 1: [12.5, Prob. 21]

If $z = x^2 + xy + y^2 \sin\left(\frac{x}{y}\right)$, show that

$$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = 2z = x^2\frac{\partial^2 z}{\partial x^2} + 2xy\frac{\partial^2 z}{\partial x\,\partial y} + y^2\frac{\partial^2 z}{\partial y^2}$$

Problem 2: [12.5, Prob. 27]

A function is said to be a harmonic function in a region R if it satisfies the Laplace's equation in $\mathbb{R}$ and has continuous second partial derivatives in $\mathbb{R}$. The Laplace's equation for a function $fx, y, z)$ of three variables is

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 0$$

Find a region (if possible) in which the function,

$$f(x, y, z) = \frac{1}{\sqrt{x^2 + y^2 + z^2}}$$

is harmonic.

Problem 3: [12.5, Prob. 49]

The Laplace's equation for a function $u(x, y)$ of two variables is

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

a. Show that the following function, is harmonic in the entire xy-plane:

$$u(x, y) = e^x \cos y + x.$$

b. In complex variable theory, two functions $u(x, y)$ and $v(x, y)$ are said to be *harmonic conjugates* in a region R if in R they satisfy the Cauchy-Riemann equations:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$$

Find a function $v(x, y)$ so that the function $u(x, y)$ and $v(x, y)$ are harmonic conjugates.

Chain Rule for Partial Derivatives (S.12.6.)

Problem 1: [12.6, Prob. 13]

Find the derivative below:

$$\left.\frac{\partial^2 u}{\partial s^2}\right|_t$$

If $u = x^2 + y^2 + z^2 + xyz,\ x = s^2 + t^2,\ y = s^2 - t^2,$ and $z = st$

Problem 2: [12.6, Prob. 27]

If $f(s)$ is a differentiable function, show that $u(x, y) = f(4x - 3y) + 5(y - x)$, satisfies the equation

$$3\,\frac{\partial u}{\partial x} + 4\,\frac{\partial u}{\partial y} = 5$$

Problem 3: [12.6, Prob. 35]

An observer travels along the curve $x = t^2,\ y = 3t^3 + 1, z = 2t + 5$, where x, y and z are in meters and $t \geq 0$ is in seconds.

If the density ρ of a gas (in kg/m^3) is given by

$$\rho = \frac{(3x^2 + y^2)}{z^2 + 5}$$

Find the time rate of change of the density of the gas as measured by the observer when $t = 2$ seconds.

Directional Derivatives (S.12.8)

Problem 1: [12.8, Prob. 9]

Find the rate of change of $f(x, y) = 2x - 3y$ at (1,1) along the curve $y = x^2$, in the direction of increasing x.

Problem 2: [12.8, Prob. 15]

Find the direction in which the function below increases most rapidly at the point $(1, -3, 2)$. What is the rate of change in that direction?

$$f(x, y, z) = \frac{1}{\sqrt{x^2 + y^2 + z^2}}$$

Problem 3: [12.8, Prob. 25]

Find points on the curve given by

$$C: x = t, y = 1 - 2t, z = t$$

at which the rate of change of the function

$$f(x, y, z) = x^2 + xyz$$

with respect to the distance travelled along the curve vanishes.

More practice problems[1]

The answers to these problems, along with all even-numbered exercises, are provided at the back of the textbook. Detailed solutions can be found in Trim's Student Solution Manual. If you don't own a copy, a reserved edition is available at the Davis Centre.

Warm-Up Problems

1. S. 12.5, Probs. 12, 16
2. S. 12.6, Probs. 2, 4, 12
3. S. 12.8, Probs. 2, 4, 10, 14

Extra Practice Problems

1. S. 12.5, Probs. 22, 30
2. S. 12.6, Probs. 18, 26
3. S. 12.8, Probs. 18, 20, 30

Extra Challenge Problems

1. S. 12.5, Prob. 36
2. S. 12.6, Prob. 44
3. S. 12.8, Prob. 34

[1] You are strongly encouraged to try and solve the problems assigned in this homework on your own, before looking at the solutions. These problems are only the minimum necessary to master the material. You are also encouraged to do additional problems from the text for practice (some suggested **extra practice problems** have been listed here).